# SS3 to opal: IO yellowfin

## Converting and fitting the IATTC sample_tree/4 operating model on both biology routes

Nicholas Ducharme-Barth

2026-10-13

## Table of contents

1 Overview ..... 1
1.1 The model ..... 2
1.2 The two biology routes ..... 2
1.3 Fit configuration ..... 3
2 Conversion checks ..... 3
3 Optimisation ..... 4
4 Index fits ..... 5
5 Composition fits ..... 6
5.1 Aggregated over time ..... 7
5.2 Selected observations ..... 7
5.3 Mean length over time ..... 9
6 Biomass trends ..... 10
6.1 Comparison with the simulated truth ..... 11
7 Depletion trends ..... 12
8 Why the fitted opal models differ from SS3 ..... 12
8.1 Test: opal with SS3′s length-at-age ..... 14
9 Differences between opal and SS3 ..... 17
9.1 Bridging check: the catch-timing effect, isolated ..... 18
9.2 What the control route costs on top ..... 18
10 Session ..... 19

## 1 Overview

This vignette documents the conversion of the IO yellowfin simulated SS3 model (`model-files/ss3/xx-io-yft-sim-iattc-sample-tree-4`, SS3 v3.24) into an `opal_obj`, and compares opal fits on both biology routes against the SS3 solution.

**Key results.**

- The conversion is exact or reconciled for catch, the index, length compositions and biology.
- At SS3′s parameter values the opal model tracks SS3′s spawning biomass within 10%. All of that remaining gap is explained by opal removing catch at the start of each timestep.
- Fitted, both opal paths converge cleanly, starting from the SS3 optimum, in about 4.4 to 5.3 minutes. They end well below SS3′s biomass, though.
- Diagnostic fits trace that gap to the length compositions as predicted through opal's single-k growth curve, which cannot represent SS3′s age-specific K. Fitted to the index alone, opal reproduces SS3′s depletion.

- Against the simulated truth (the operating model the data were simulated from), SS3 itself overestimates B0 by 19% and ends at a depletion of 0.399, against a true 0.468. The opal fits are compared with the truth in section 6.1.
- A test with a local opal patch confirms it. Give opal SS3′s own start-of-timestep length-at-age and fit with the length compositions on: terminal SSB is then within 4% of SS3, and terminal depletion is 0.404 against SS3′s 0.399 (section 8.1).

The workflow has three steps. Only the first two touch SS3 or run an optimiser; this document reads a cache and rebuilds in seconds.

| Step | File | Does |
|---|---|---|
| 1 | `code/rtmb/scripts/00-prep-io-yft-sim-data-for-rtmb.r` | Convert, validate, reconcile, build and save the unfitted `opal_obj` |
| 2 | `code/rtmb/scripts/01-fit-io-yft-ss3-opal.r` | Fit both biology routes, time them, write `vignettes/data/io-yft-ss3-opal-cache.rds` |
| 3 | `code/rtmb/scripts/02-extract-io-yft-sim-truth.r` | Sum the operating model's truth over regions, write `vignettes/data/io-yft-sim-truth-cache.rds` |
| 4 | `code/rtmb/vignettes/ss3-to-opal-io-yft.qmd` | This document; reads only the two caches |

To rebuild after a change to the converter, the config or the SS3 files, rerun step 2 (step 3 only if the truth files change) and then `quarto render code/rtmb/vignettes/ss3-to-opal-io-yft.qmd`. The cache here was written 2026-10-13 13:24:03 UTC with R 4.5.3, opal 0.0.4 and r4ss 1.56.0.

## 1.1 The model

SS3 runs 256 "years" of three months each (`nseas = 1`, `months_per_seas = 3`), so each SS3 year is a quarter. opal takes `n_year = 256` and `n_season = 1`. There are 16 extraction fleets with catch in numbers and one longline CPUE index. Following the original prep script, the index is split into four quarterly series, each with its own catchability (opal fisheries 17 to 20). SS3 ages 1 to 28 are carried, so the opal age index equals the SS3 age.

The data were simulated from replicate 4 of the Indian Ocean yellowfin operating model of the Spatial Stock Assessment Simulation Experiment: a spatial population model with 4 regions, also run in quarterly "years" 1–256. Its spawning biomass is mature biomass at the start of each year, before fishing, in tonnes. That is the same timing and units as SS3′s, so the two can share an axis once the truth is summed over regions. True B0 is the operating model's unfished equilibrium SSB, 3,928,880 t. Definitions and file provenance are in `model-files/sim-truth/io-yft-iattc-sample-tree-4/README.md`.

## 1.2 The two biology routes

| | endgrowth | control |
|---|---|---|
| Source | SS3′s `endgrowth` table in Report.sso | Control-file parameters |
| M-at-age | endgrowth M / 4 | control natM / 4 |
| Weight-at-age | SS3 `Wt_Beg`, age basis | length-weight at bin midpoints, mapped to age by opal's PLA |
| Maturity | SS3 `Age_Mat`, age basis | control `Age_Maturity`, age basis |

| | endgrowth | control |
|---|---|---|
| Spawning potential | SS3 `Mat*Fecund`, supplied directly | maturity × fecundity-at-age, via the PLA |
| Growth used for | the PLA behind predicted length compositions | the PLA behind compositions **and** weight / spawning potential |

On both routes the opal growth curve is a single-k Schnute curve. SS3 uses `GrowthModel 3`, with age-specific K multipliers over ages 2 to 10, which that curve cannot reproduce. The endgrowth route keeps the error out of biomass; the control route does not.

### 1.3 Fit configuration

Both routes are fitted in an ASPM-R configuration. SS3's selectivity-at-age (`Asel2`, or `Asel` for the survey) is injected as `sel_fa_external` and held fixed. opal estimates `log_B0`, the 256 recruitment deviations and the four quarterly catchabilities: 261 parameters. Steepness, σ_R and growth are fixed at their SS3 values. Composition likelihoods are multinomial on fisheries 1–14 and 16, with SS3's `mult_by_lencomp_N` carried as opal's `lf_var_adjust`.

## 2 Conversion checks

The converter reconciles its output against SS3 before anything is fitted. Rows with `pass = NA` are known approximations, reported rather than tested.

| quantity | ss3 | opal | rel_diff | tol | pass |
|---|---|---|---|---|---|
| catch total, fleet 1 (fishing_gi_1) | 58644.84000 | 58644.84000 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 2 (fishing_gi_4) | 19412.76000 | 19412.76000 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 3 (fishing_hd_1) | 47115.22000 | 47115.22000 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 4 (fishing_ll_1) | 47074.37000 | 47074.37000 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 5 (fishing_ll_2) | 9146.36500 | 9146.36500 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 6 (fishing_ll_3) | 4744.18700 | 4744.18700 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 7 (fishing_ll_4) | 33189.32000 | 33189.32000 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 8 (fishing_other_1) | 58433.02000 | 58433.02000 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 9 (fishing_other_4) | 40323.64000 | 40323.64000 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 10 (fishing_bb_1) | 138227.80000 | 138227.80000 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 11 (fishing_ps_1) | 369226.90000 | 369226.90000 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 12 (fishing_ps_2) | 30289.96000 | 30289.96000 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 13 (fishing_ps_4) | 9416.36500 | 9416.36500 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 14 (fishing_trol_1) | 37630.63000 | 37630.63000 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 15 (fishing_trol_2) | 32907.52000 | 32907.52000 | 0.00e+00 | 1e-06 | TRUE |
| catch total, fleet 16 (fishing_trol_4) | 92779.85000 | 92779.85000 | 0.00e+00 | 1e-06 | TRUE |
| CPUE observation count | 176.00000 | 176.00000 | 0.00e+00 | 1e-06 | TRUE |
| LF total input sample size | 5210.00000 | 5210.00000 | 0.00e+00 | 1e-06 | TRUE |
| LF observation rows | 1114.00000 | 1114.00000 | 0.00e+00 | 1e-06 | TRUE |
| unfished spawning potential per recruit | 39.99116 | 39.99303 | 4.69e-05 | 1e-04 | TRUE |

| quantity | ss3 | opal | rel_diff | tol | pass |
|---|---|---|---|---|---|
| total weight summed over ages | 939.48810 | 939.48810 | 0.00e+00 | 1e-06 | TRUE |
| mean length-at-age, Schnute approximation | NA | NA | 7.63e-02 | | NA |
| SD of length-at-age, CV approximation | NA | NA | 6.12e-02 | | NA |

The two non-trivial rows:

- **Unfished spawning potential per recruit** agrees to 4.7e-05. This one number tests the units of M, the female fraction and the plus group together. The small remaining difference is real. SS3 3.24 builds the plus group by iterating the equilibrium out to 3 × Nages and closes the tail with the annual Z. opal uses the closed-form geometric tail at the per-timestep M.
- **Growth.** The Schnute fit to SS3's length-at-age has an RMSE of 7.86 cm (endgrowth route, fitted k). The control-file curve has 12.02 cm, with a maximum error of 28.59 cm.

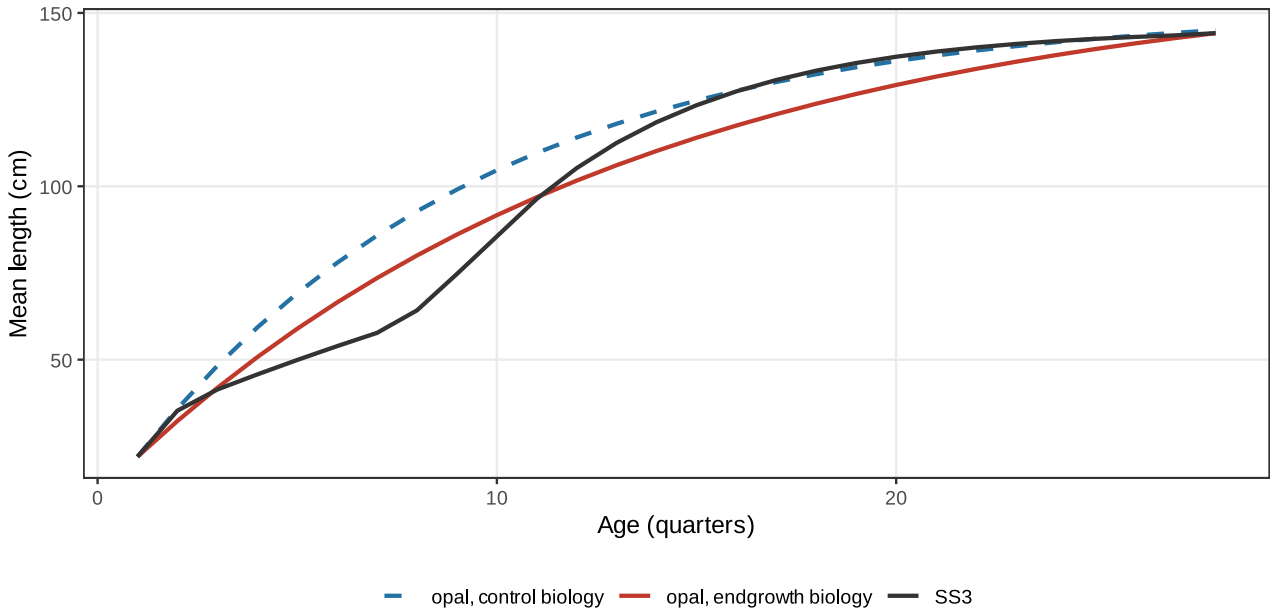



Figure 1: Mean length-at-age: SS3 (age-specific K) against the two opal Schnute curves.

# 3 Optimisation

Every fit starts from the converted SS3 optimum: SS3's unfished SSB as `B0`, SS3's estimated recruitment deviations, and q = 1 on the centred index scale. **These timings are for refining a solution that is already close, not for finding one from default starting values.** A cold start would take longer.

| Stage | endgrowth (s) | control (s) |
|---|---|---|
| convert SS3 files | 14.9 | 10.7 |
| build opal objects (two tapes) | 60.2 | 60.8 |
| opal_fit: tape + nlminb (2 passes) | 67.0 | 110.8 |
| rebuild tape | 26.3 | 27.3 |
| Newton polish | 26.5 | 29.3 |
| attach fit + opal_check | 70.8 | 78.4 |
| total | 265.7 | 317.2 |

| route | pass | objective_start | objective | convergence | iterations | fn_evals | gr_evals |
|---|---|---|---|---|---|---|---|
| control | 1 | 22198.61 | 21394.53 | 0 | 985 | 1112 | 985 |
| control | 2 | 22198.61 | 21394.53 | 0 | 130 | 174 | 131 |
| end-growth | 1 | 17658.24 | 17519.18 | 0 | 318 | 372 | 319 |
| end-growth | 2 | 17658.24 | 17519.18 | 0 | 33 | 38 | 34 |

| Route | Para-meters | Newton steps | Max \|grad\|, nlminb | Max \|grad\|, final | opal check | Estimable |
|---|---|---|---|---|---|---|
| endgrowth | 261 | 2 | 2.5e-03 | 7.6e-12 | passed | all |
| con-trol | 261 | 2 | 1.4e-02 | 5.2e-10 | passed | all |

Taping dominates the wall time. opal's objective is built by RTMB over 256 quarterly timesteps, 28 ages and 20 fisheries, and `opal_fit()` rebuilds the tape before optimising. nlminb stops on relative function change, which leaves a maximum gradient of order 1e-3 on an objective near $1.7 \times 10^4$. Two Newton steps then bring it below 1e-8, and opal's own fit check passes at a gradient tolerance of 1e-4. Both routes are estimable: positive-definite Hessian, and all 261 parameters identified. The control route takes more nlminb iterations because it starts further from its optimum: its biology moves R0 away from SS3's even at SS3's own B0.

The diagnostic fits in section 8 took 158 s (no length compositions) and 350 s (growth estimated) end to end. The SS3 length-at-age tests took 218 s (start of timestep) and 219 s (mid-timestep).

For reference, the SS3 Report.sso header records a run time of 50.4 minutes for 315 active parameters (SS3 also estimates selectivity and its own catchability). That run started from SS3's own initial values, and the header does not say whether a Hessian was included or on what hardware, so it is context, not a benchmark. The opal timings are from a Intel(R) Xeon(R) Processor @ 2.80GHz machine with 4 cores. Treat them as indicative. The same script, rerun on this cloud container after it was restarted, took 1.5 to 2 times as long at every stage, including reading the SS3 files, with identical iteration and evaluation counts. The iteration counts above are the reproducible measure of optimisation effort.

# 4 Index fits

SS3 fits the longline index as one series with one catchability, and samples it mid-season (`surveytiming = 0.5`). opal fits four quarterly series, each with its own catchability, against abundance at the start of the timestep. Both indices are shown on a mean-one scale per quarterly series.

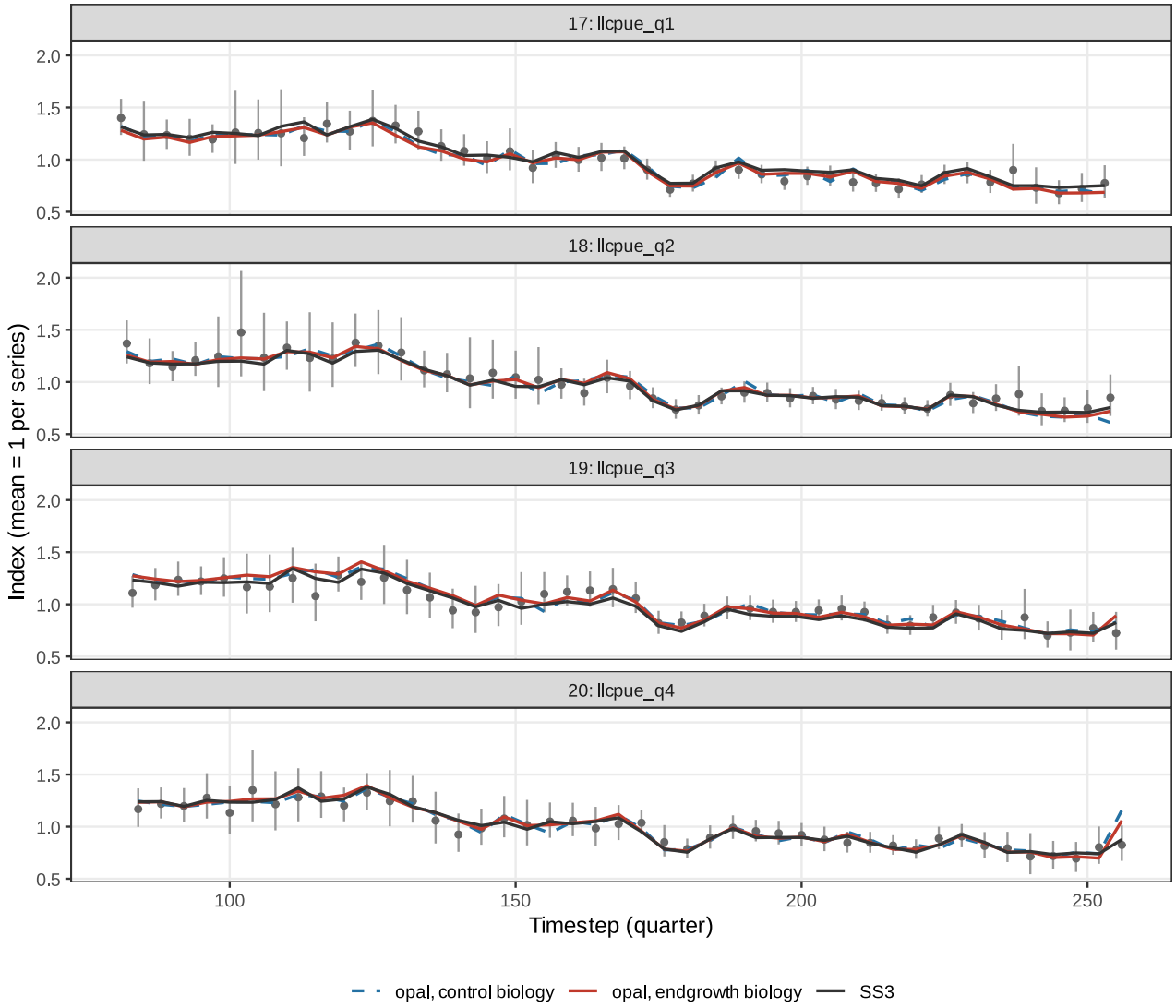



Figure 2: Observed index (points, ±1.96 SE) with the fitted opal predictions on both routes and the SS3 expected values.

| Model | RMSE of log residuals |
|---|---|
| opal, endgrowth biology | 0.0695 |
| opal, control biology | 0.0778 |
| SS3 | 0.0646 |

# 5 Composition fits

opal predicts length compositions by passing catch-at-age through its probability-of-length-at-age matrix (PLA), which is built from the Schnute growth curve. SS3 uses its own age-length key. Selectivity is SS3's on all three models, so composition differences come from growth, from population numbers-at-age, and from the timing of the catch within the timestep. SS3's expected compositions are taken from the expected-value section of `data.ss_new`, because CompReport.sso was not written for this run.

## 5.1 Aggregated over time

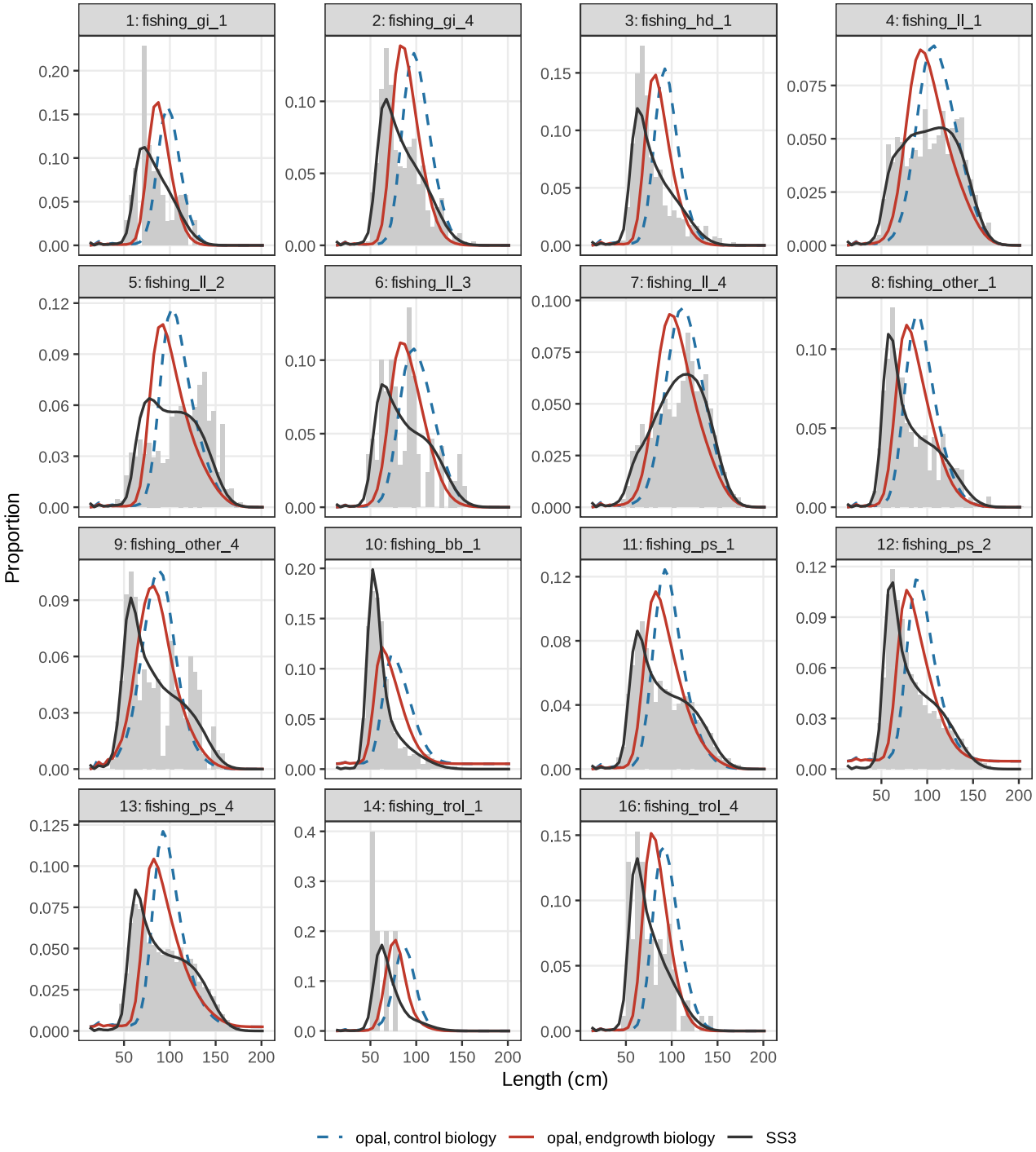



Figure 3: Length compositions aggregated over all observations for each fishery, weighted by effective sample size.

## 5.2 Selected observations

The four observations with the largest effective sample size, from four fisheries with different selectivity shapes.

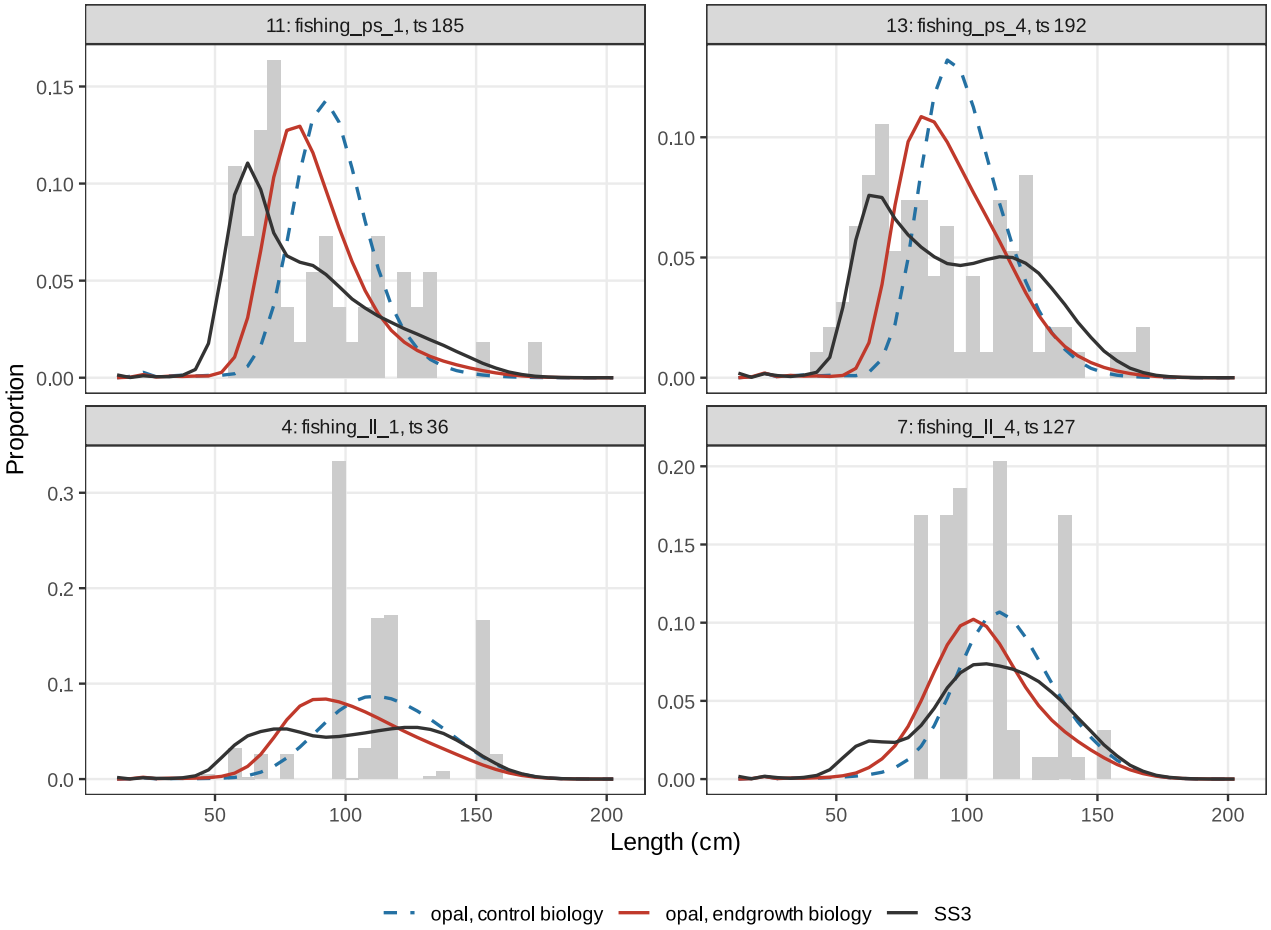



Figure 4: Individual length composition observations (bars) with predictions.

## 5.3 Mean length over time

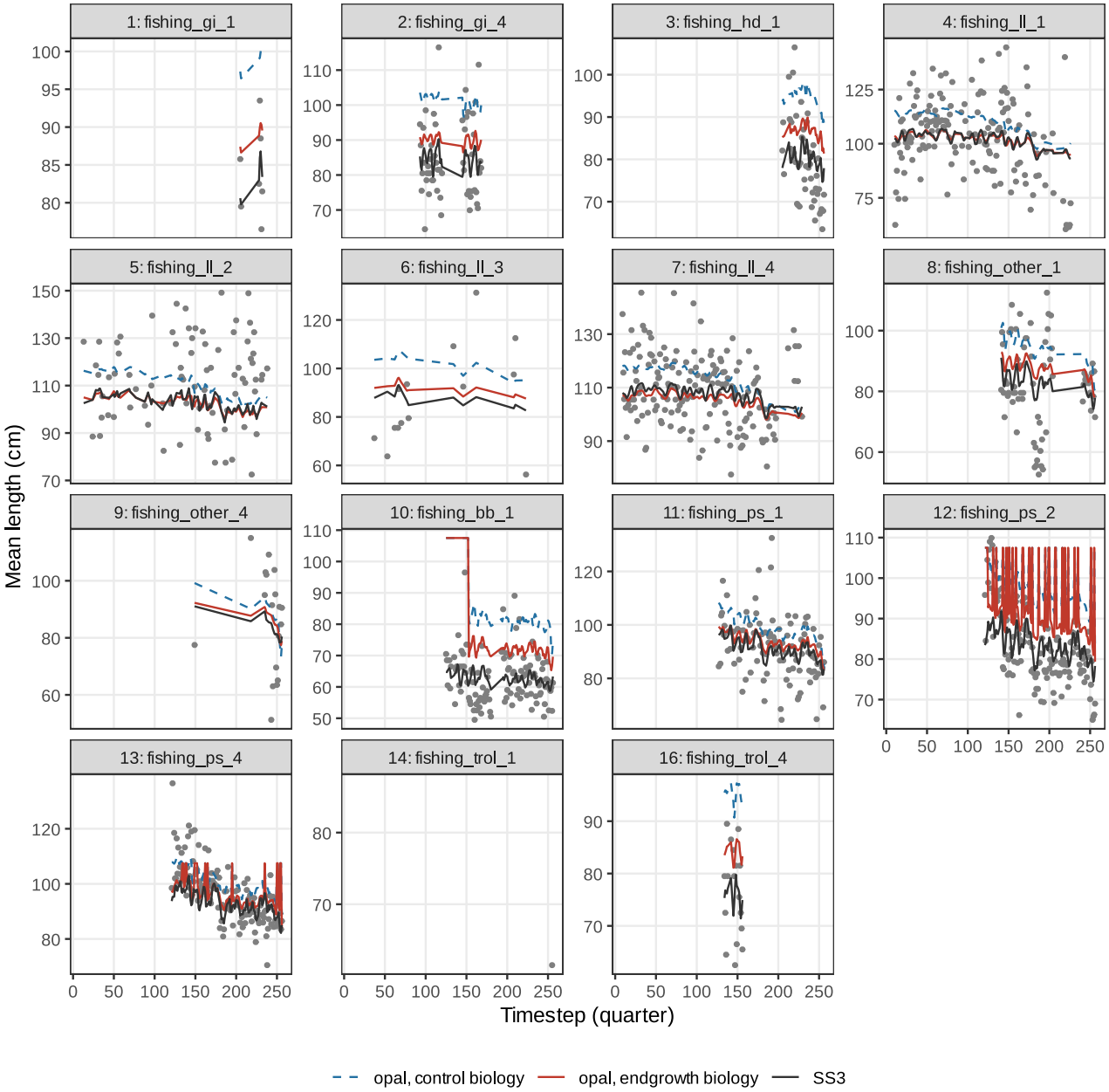



Figure 5: Observed and predicted mean length per observation.

# 6 Biomass trends

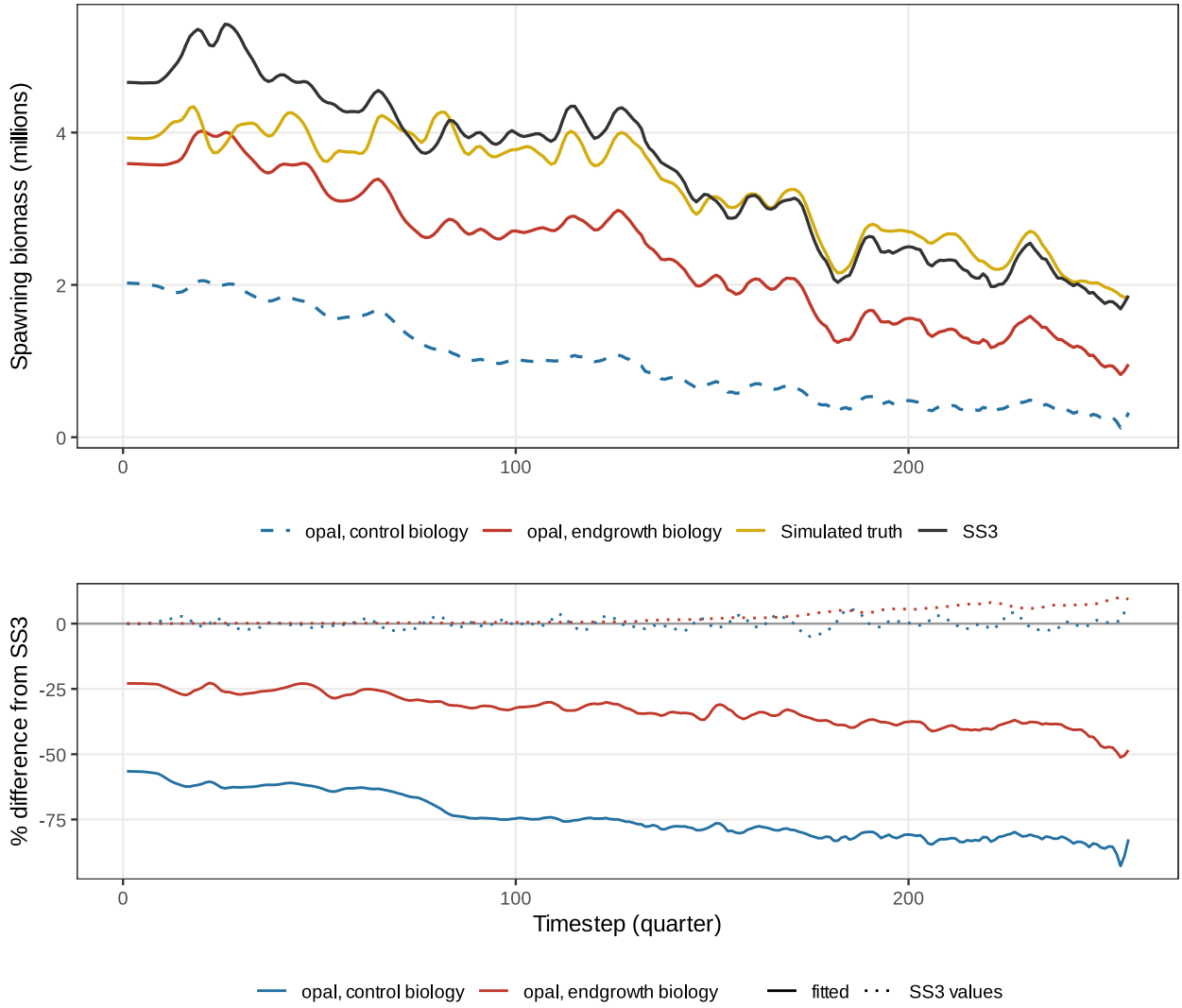



Figure 6: Spawning biomass. Top: the simulated truth, SS3 and the fitted opal models. Bottom: difference from SS3, fitted (solid) and at the SS3 parameter values before fitting (dotted).

| Model | Parameters | max \|% diff\| | mean \|% diff\| | terminal % diff |
|---|---|---|---|---|
| opal, endgrowth biology | SS3 values | 10.28 | 2.54 | 9.35 |
| opal, endgrowth biology | fitted | 51.23 | 32.97 | −48.44 |
| opal, control biology | SS3 values | 6.25 | 1.46 | 6.25 |
| opal, control biology | fitted | 92.81 | 73.98 | −82.62 |

| route | B0_start | B0_fit | R0_start | R0_fit |
|---|---|---|---|---|
| endgrowth | 4,659,770 | 3,592,748 | 116,515 | 89,834 |
| control | 4,659,770 | 2,025,039 | 99,657 | 43,309 |

SS3's unfished spawning biomass is 4,659,770 and its R0 is 116,520.

## 6.1 Comparison with the simulated truth

Every model is compared here with the operating model, including the diagnostic fits from section 8. Terminal values are for the last quarter, 256.

| Model | B0 (Mt) | B0 / true B0 | SSB mean \|% diff\| | SSB terminal % diff | Terminal SSB/B0 | Max \|depletion error\| |
|---|---|---|---|---|---|---|
| SS3 | 4.66 | 1.19 | 9.8 | 1.0 | 0.399 | 0.252 |
| opal, endgrowth biology | 3.59 | 0.91 | 29.9 | −47.9 | 0.267 | 0.326 |
| opal, control biology | 2.03 | 0.52 | 72.1 | −82.4 | 0.160 | 0.524 |
| opal, endgrowth, no length comps | 4.72 | 1.20 | 8.7 | 2.4 | 0.399 | 0.222 |
| opal, endgrowth, growth estimated | 9.39 | 2.39 | 138.5 | 169.5 | 0.528 | 0.194 |
| opal, endgrowth, SS3 length-at-age (start) | 4.78 | 1.22 | 11.0 | 5.1 | 0.404 | 0.214 |
| opal, endgrowth, SS3 length-at-age (mid) | 4.22 | 1.07 | 12.1 | −15.7 | 0.367 | 0.235 |
| Simulated truth | 3.93 | 1.00 | NA | NA | 0.468 | NA |

SS3 is not the truth either. Its terminal SSB is within 1% of the truth, but its B0 is 1.19 × the operating model's, so it ends more depleted (0.399 against 0.468). Its depletion error comes from the unfished reference point, not the current stock size. Part of the difference is structural: the truth has 4 regions with movement and maturity as a process, while SS3 is a single-region model with an age-based maturity ogive. So "matches SS3" and "matches the truth" are separate questions. The bridging work in this vignette answers the first. The table answers the second, for SS3 and for each opal configuration.

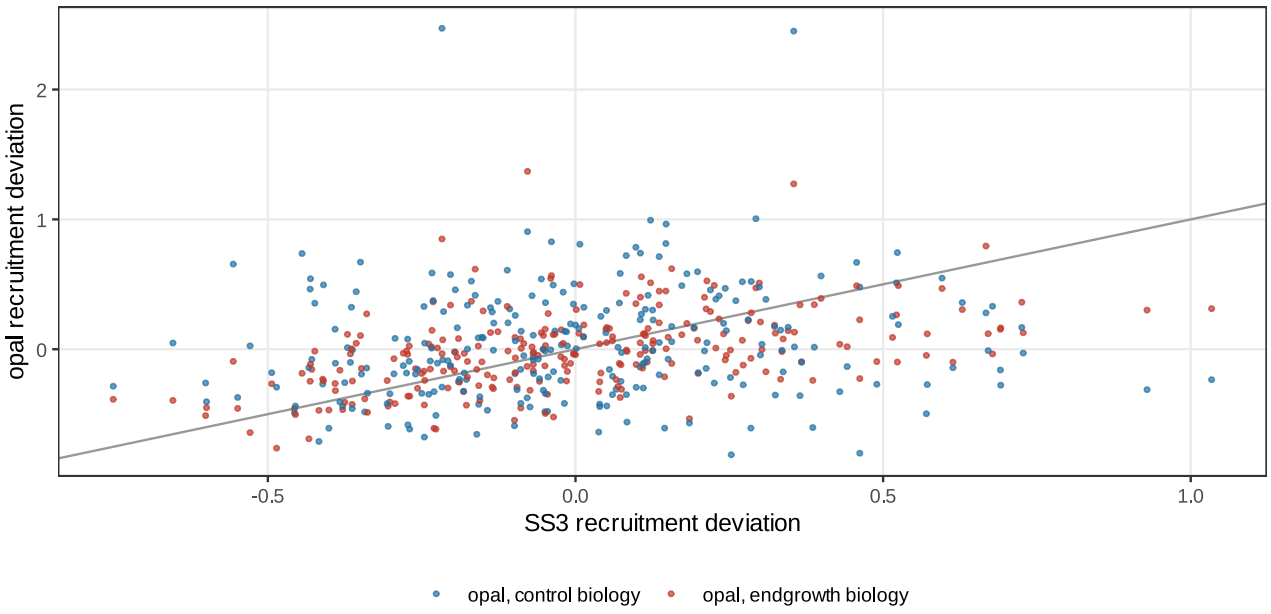



Figure 7: Estimated recruitment deviations against SS3's. opal's deviation for timestep t sets the age-1 recruits entering at t + 1.

# 7 Depletion trends

Static depletion is SSB / B0. opal also reports dynamic depletion, the ratio of SSB to the unfished trajectory under the same recruitment deviations. SS3 3.24 does not report the dynamic version, so only static depletion is compared with SS3.




Figure 8: Static depletion (SSB / B0) for the simulated truth, SS3 and both fitted opal routes, with opal's dynamic depletion (lighter).

| Model | terminal opal | terminal SS3 | terminal truth | max \|diff\| from SS3 | max \|diff\| from truth |
|---|---|---|---|---|---|
| opal, endgrowth biology | 0.267 | 0.399 | 0.468 | 0.135 | 0.326 |
| opal, control biology | 0.160 | 0.399 | 0.468 | 0.412 | 0.524 |

# 8 Why the fitted opal models differ from SS3

The fitted opal trajectories end well below SS3: by 48% on the endgrowth route and 83% on the control route. At SS3's own parameter values, before fitting, the endgrowth route is within 10% of SS3 throughout. So the fit moves away from the SS3 solution rather than sitting on it. Two diagnostic fits on the endgrowth route locate the cause:

- **No length compositions.** The same model, with the composition likelihood switched off, so only the index and the recruitment penalty inform the fit.
- **Growth estimated.** The composition likelihood is on, and L1, L2, k, CV1 and CV2 are estimated (opal's default priors apply). This lets the PLA, which maps catch-at-age to predicted lengths, adapt.

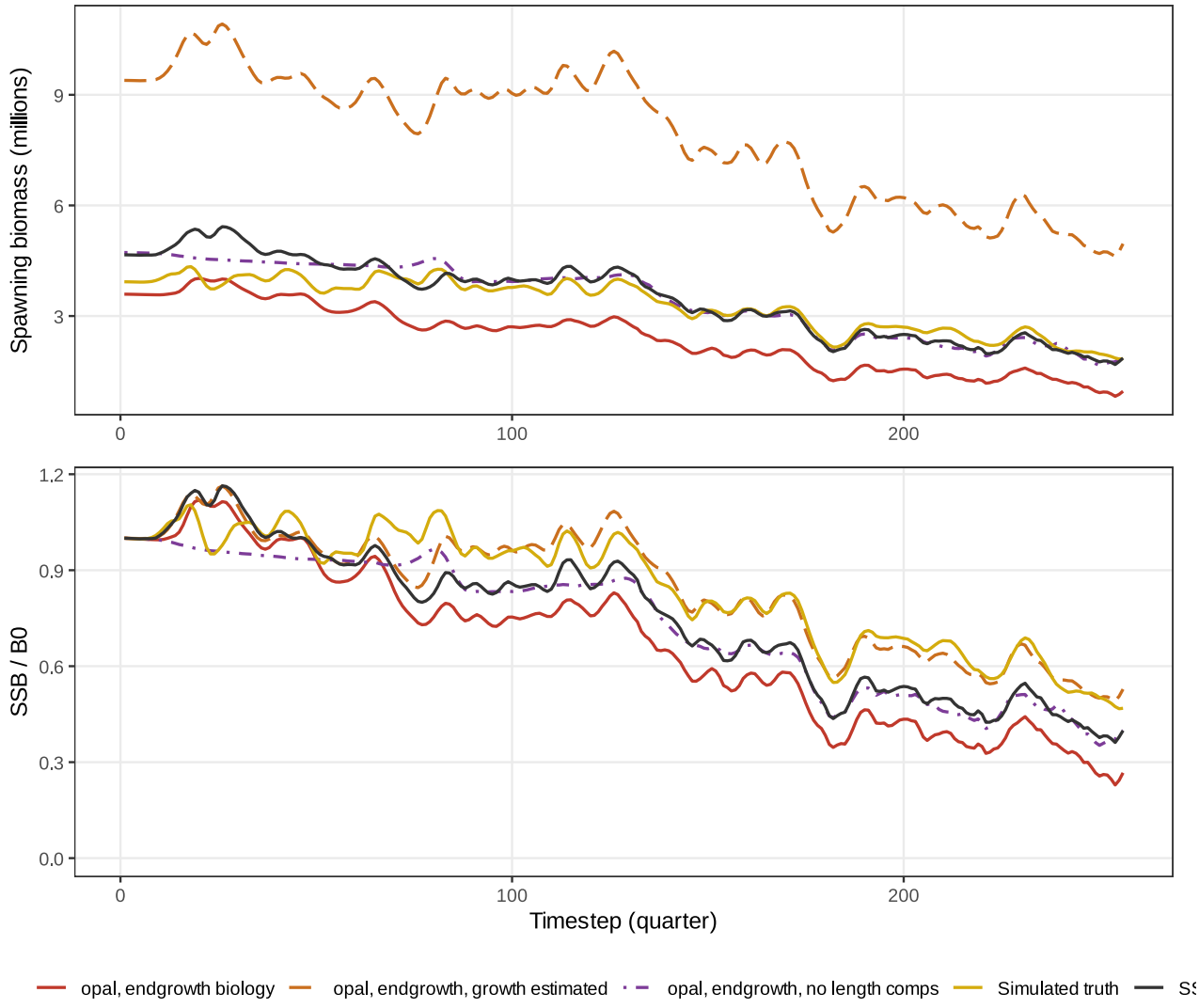



Figure 9: Spawning biomass (top) and static depletion (bottom) for the simulated truth, SS3, the endgrowth fit and the two diagnostic fits.

| Model | max \|% diff\| SSB | mean \|% diff\| SSB | terminal % diff SSB | terminal SSB/B0 | B0 | R0 | NLL index | NLL length comps |
|---|---|---|---|---|---|---|---|---|
| opal, endgrowth biology | 51.2 | 33.0 | −48.4 | 0.26 | 57 | 92,748 | 89,834–229.0 | 17,567.1 |
| opal, control biology | 92.8 | 74.0 | −82.6 | 0.12 | 60 | 25,034 | 93,309–211.5 | 21,394.9 |
| opal, endgrowth, no length comps | 20.3 | 4.0 | 1.4 | 0.39 | 97 | 22,781 | 18,090–251.3 | 0.0 |
| opal, endgrowth, growth estimated | 174.1 | 131.3 | 166.9 | 0.92 | 38 | 91,923 | 94,839–238.2 | 14,783.7 |
| opal, endgrowth, SS3 length-at-age (start) | 7.9 | 4.6 | 4.0 | 0.41 | 78 | 80,127 | 79,524–240.2 | 14,253.9 |
| opal, endgrowth, SS3 length-at-age (mid) | 17.5 | 11.3 | −16.5 | 0.36 | 72 | 21,770 | 05,563–239.7 | 14,243.1 |

| Model | max \|% diff\| SSB | mean \|% diff\| SSB | terminal % diff SSB | terminal SSB/B0 | B0 | R0 | NLL index | NLL length comps |
|---|---|---|---|---|---|---|---|---|
| SS3 | NA | NA | NA | 0.399 | 59,770 | 6,520 | NA | NA |

Without the length compositions, opal lands on SS3. Terminal depletion is 0.399, against SS3's 0.399, and terminal SSB is within 1.4%. On average it is within 4.0%, although the single worst timestep is still 20% off, because the index alone constrains individual recruitment deviations only weakly. The remaining differences are the catch-timing and spawner-recruit-timing effects described below. **The divergence of the fitted models therefore comes from the length compositions**, and specifically from how opal predicts them.

opal predicts a length composition as catch-at-age multiplied by the PLA built from its Schnute curve. Selectivity here is SS3's age-based curve, so the PLA is the only route from ages to lengths. The fitted Schnute curve is 16 cm too long at age 8 and 5 cm too long at age 4, exactly where most of the catch is taken. It cannot bend the way SS3's age-specific K does. Predicted compositions therefore sit to the right of the data on every fishery (section 5). The optimiser can only answer with population structure: fewer old fish, lower recruitment and a smaller stock. On the control route, the same curve also inflates weight-at-age, and the scale moves further.

Freeing growth confirms that the compositions dominate. They pull the curve down to L1 = 10.0 cm at age 1 (from 22 cm), despite the prior. The composition NLL falls by 2,783 units, and B0 then overshoots to 2.0 × SS3's. A single-k curve fitted to the compositions is still not SS3's growth, so this is not a fix. It shows how much leverage the compositions have over scale when growth is misspecified.

## 8.1 Test: opal with SS3's length-at-age

If growth is the cause, giving opal SS3's own mean and SD of length-at-age should bring the fitted model back to SS3, with the length compositions still in the likelihood. opal has no option for this, so the test uses a **local patch**. `ss3_patch_opal_external_laa()` (in `ss3-opal-obj.R`) inserts one statement into opal's `opal_model()` for the current R session, after it computes length-at-age:

```
if (exists("laa_mu_external", inherits = FALSE)) {
    mu_a <- laa_mu_external
    sd_a <- laa_sd_external
}
```

The installed opal is not changed. The patch does nothing for a model without those two data vectors, and objects built under it carry a different model signature from stock opal. Everything else matches the endgrowth fit: biology, SS3's selectivity, the same starting point and the same estimated parameters. Two versions of SS3's length-at-age are tried. `Len_Beg`/`SD_Beg` are lengths at the start of the timestep, which is when opal removes the catch. `Len_Mid`/`SD_Mid` are lengths at mid-timestep, which SS3 uses for its own catch compositions.

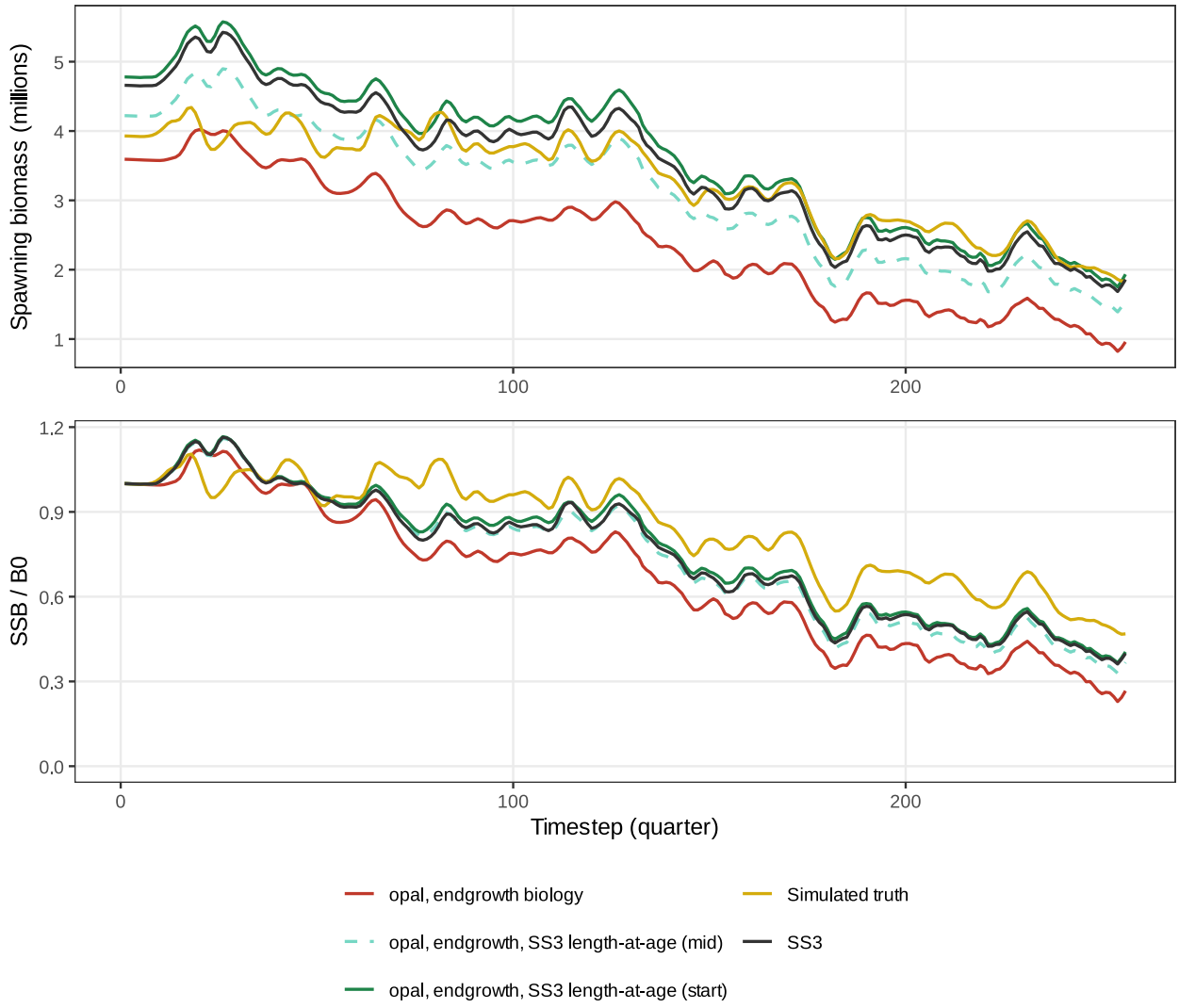



Figure 10: Spawning biomass (top) and static depletion (bottom): the simulated truth, SS3, the stock endgrowth fit, and the endgrowth fit with SS3's length-at-age.

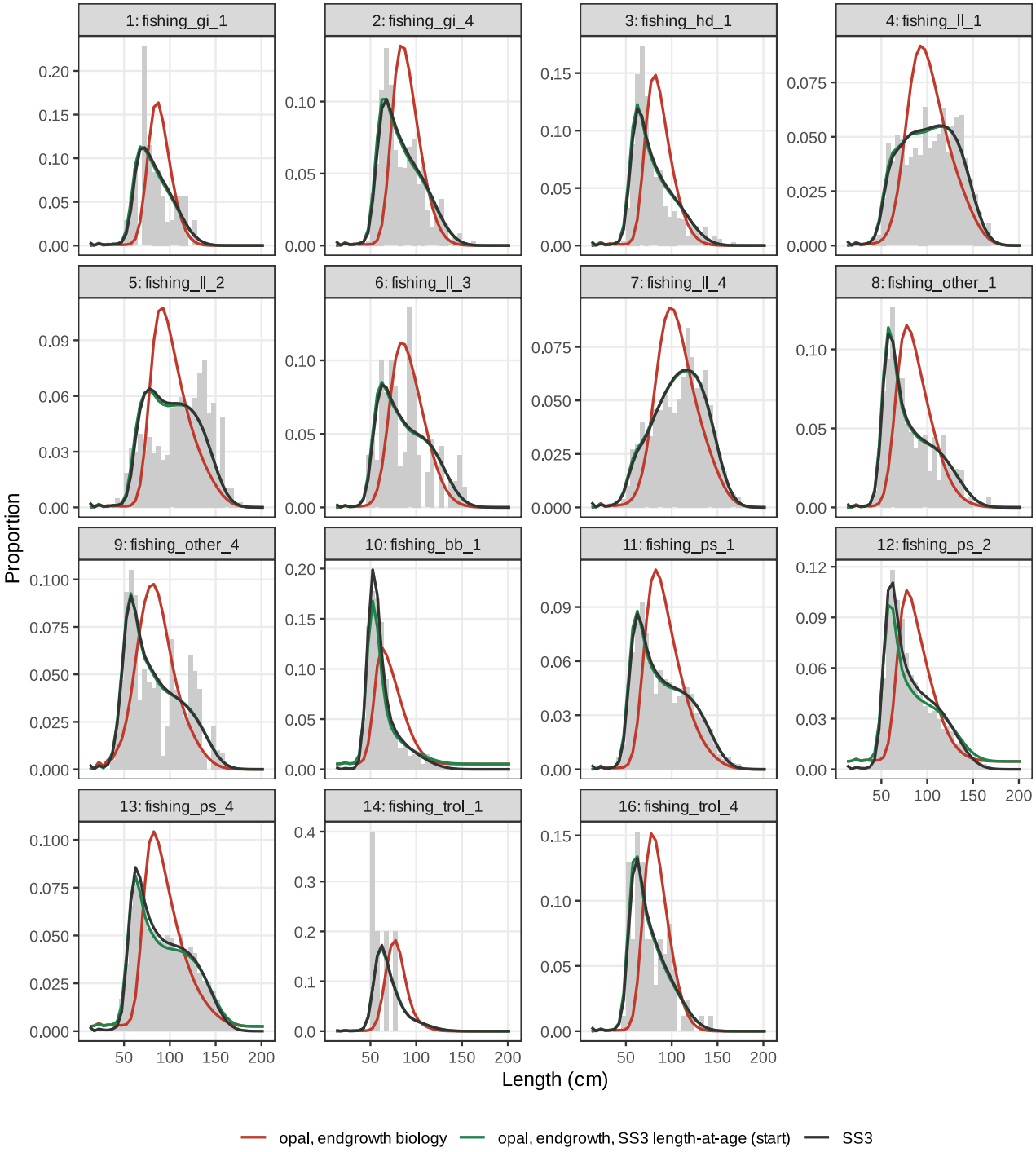



Figure 11: Aggregated length compositions: the stock endgrowth fit against the fit with SS3′s start-of-timestep length-at-age, and SS3′s own expectation.

With SS3′s start-of-timestep length-at-age, the predicted compositions sit on SS3′s, and the fitted model returns to SS3. B0 is 1.03 × SS3′s. Mean and maximum SSB differences are 4.6% and 7.9%, comparable to the catch-timing effect measured in the bridging check (up to 10% at SS3′s parameter values). The composition NLL falls from 17,567 to 14,254, lower than the growth-estimated fit reaches (14,784). **The growth explanation holds: with SS3′s growth, opal fits the compositions better than the growth-estimated single-k fit does, and lands on SS3′s stock size.**

The mid-timestep version lands 17% below SS3 at the end of the series, with B0 at 0.91 × SS3′s, yet its composition NLL (14,243) is almost the same as the start-of-timestep fit's. Half a quarter of growth moves the estimated scale by about 20% without changing the fit to the compositions. Two lessons follow:

- The start-of-timestep lengths are the consistent choice for opal, because opal removes the catch at the start of the timestep. SS3 uses mid-season lengths because its catch is spread through the season.
- Scale information from these compositions is weak. Small, defensible choices in how lengths are assigned to the catch move B0 a long way. That is worth remembering whenever opal's growth is estimated or approximated.

The practical conclusions:

- For a bridging comparison against SS3, the converted model is right. It reproduces SS3 at SS3's parameter values, when fitted to the index alone, and, with SS3's length-at-age, when fitted to everything.
- The fitted comparison in stock opal is limited by opal's growth form, not by the conversion. A data option for external length-at-age, the analogue of `sel_fa_external`, would let opal bridge to SS3 without the patch. It is a small change to `opal_model()` (one statement, as above) and the most direct fix. Age-specific k multipliers would be the route if growth is to be estimated.
- Until then, the converter's Schnute fit could weight ages by catch-at-age rather than equally, which would put the approximation error where it matters least. It is not done here.

The composition likelihood values are not comparable across models. opal's multinomial keeps the normalising constant, while SS3's omits it.

# 9 Differences between opal and SS3

The conversion itself is exact or reconciled for catch, the index, length compositions, M, weight, maturity and spawning potential. What remains are differences in model structure, listed roughly in order of their effect on the comparisons above.

| Difference | SS3 (this model) | opal | Effect | Route |
|---|---|---|---|---|
| Growth and the age-length key | GrowthModel 3, age-specific K; SS3's own ALK at mid-season | Single-k Schnute, which builds the PLA, applied at the start of the step | Predicted length compositions shift right. Through the composition likelihood this dominates the fitted difference; replacing it with SS3's length-at-age (local patch) removes it (section 8.1). On the control route it also sets weight and spawning potential | both; biology on control |
| Biology | Age-based, exact | endgrowth: SS3's age vectors; control: length-based through the PLA | Control-route R0 is −14% from SS3 even at SS3's B0 | control |
| Catch timing | Hybrid F, Baranov: F and M act together through the step (`F_Method = 3`) | Catch removed instantaneously at the start of the step, then M | opal ends each step with more fish under heavy catch, and drifts up to 10% above SS3 at SS3's parameters | both |

| Difference | SS3 (this model) | opal | Effect | Route |
|---|---|---|---|---|
| Spawner-recruit timing | Age-0 recruits from SSB at t | Age-1 recruits at t + 1 from SSB at t + 1 (SS3 age 0 is not carried) | One-step lag in the stock-recruit link. Minor, because SSB changes slowly | both |
| Index | One series, one q (analytic), sampled mid-season | Four quarterly series, four estimated q, abundance at the start of the step | Index fits differ in level by quarter | both |
| Equilibrium plus group | Iterated to 3 × Nages, tail closed with annual Z | Closed-form geometric tail | ≈5e-5 in SPR0, so in R0 for a given B0 | both |
| Priors | No prior on ln R0 | Weak normal priors on log B0 (sd 1, centred on the start) and log q (sd 1.5) | Small; B0 still moves by tens of percent | both |
| Size-frequency data | Generalised size comps for fleet 17, λ = 0.51 | Not included (`sizefreq_include = false`) | opal omits that likelihood component | both |
| Last length bin | 198+ plus bin; 195 spans 195–198 | Regular 5 cm grid; bin 200, 195 spans 195–200 | Negligible: very few fish above 195 cm | both |
| Selectivity | Estimated, age-based | SS3's selectivity-at-age, fixed | None by construction in this configuration | both |

## 9.1 Bridging check: the catch-timing effect, isolated

At SS3's parameter values (the dotted lines in the biomass figure), the endgrowth route drifts from SS3 by up to 10.3%. `ss3_bridge_replica()` re-runs the population on the same converted inputs outside opal. With opal's timing it reproduces opal's spawning biomass to about 1e-6. With Baranov catch and SS3's spawner-recruit timing, it tracks SS3 to within 0.44% at every timestep (0.09% on average). So the inputs are converted correctly, and that drift comes from where opal removes the catch. The driver writes the full comparison to `opal_data/io-yft-ss3/ioyft_bridge.csv`.

## 9.2 What the control route costs on top

The control route adds the growth approximation to biology as well as to the compositions. Fitted, the two routes differ by:

| quantity | value |
|---|---|
| SSB, max \|% diff\| control vs endgrowth | 85.259 |
| SSB, mean \|% diff\| control vs endgrowth | 61.963 |
| static depletion, max \|diff\| | 0.297 |
| R0, % diff control vs endgrowth (fitted) | −51.790 |

Use the endgrowth route to bridge to SS3, because biology is then exact. Use the control route only when growth is to be estimated, and accept that the growth curve then has to carry SS3's age-specific K pattern.

## 10 Session

| item | value |
|---|---|
| r | 4.5.3 |
| opal | 0.0.4 |
| r4ss | 1.56.0 |
| cpu | Intel(R) Xeon(R) Processor @ 2.80GHz |
| cores | 4 |